



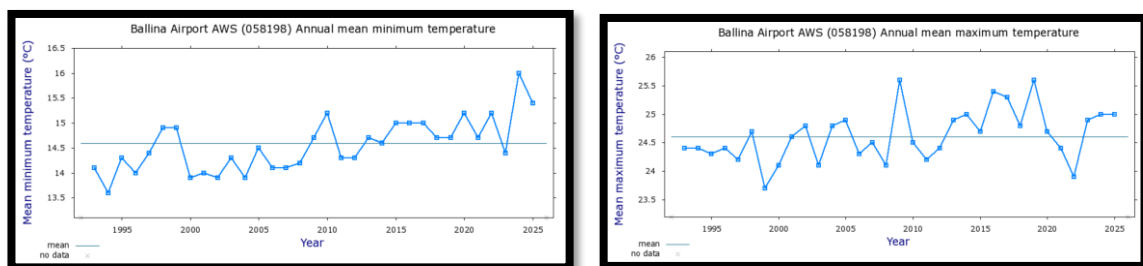
SUBMERGED: Ballina shown underwater on a predicted high tide in 2100. Photo: coastalrisk.com.au

Recent climate projections paint a concerning picture for Ballina, a coastal town that is already vulnerable to [environmental hazards](#). Important climate indicators such as temperature, precipitation, extreme fire weather occurrences and sea-level rise are all moving in directions that heighten pressure on the region. Certain changes may seem minor initially, but long-term forecasts indicate that they could have more substantial effects capable of transforming the environment and everyday life. This article examines the evolution of these trends over the past decades and their implications for Ballina's future, utilizing the most current data and climate models.

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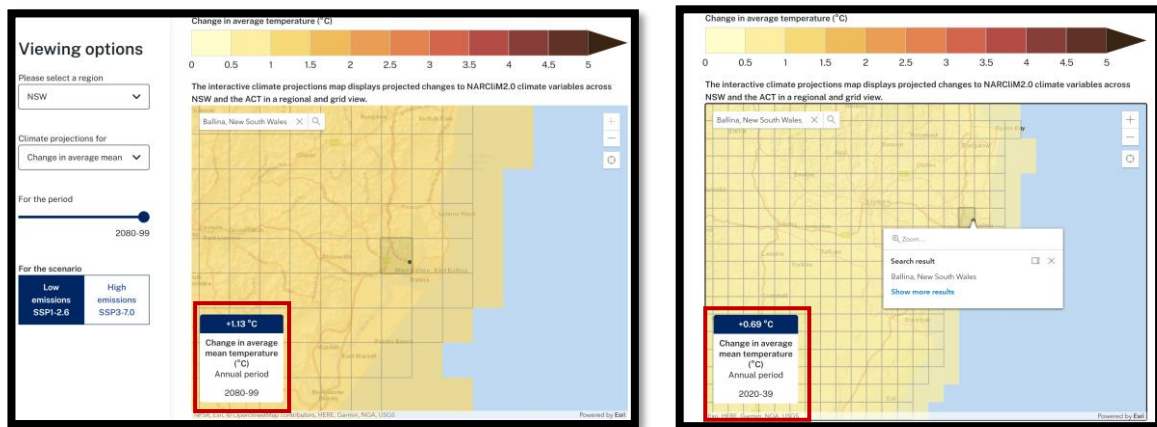
Temperature trends and projections

Temperature data from Ballina Airport indicate a gradual increase in temperature over the last few decades. Temperatures have risen for both maximum and minimum values, with nighttime temperatures rising more significantly, reducing cooling periods.



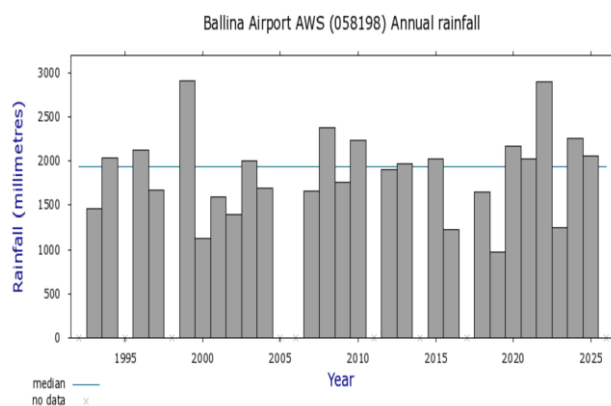
Annual mean minimum and maximum temperature trend. Photo: Bureau of methodology

Climate projections ([NARClIM2.0](#), SSP1-2.6) indicate an increase of $+0.69^{\circ}\text{C}$ in the short term (2020–2039) and $+1.13^{\circ}\text{C}$ by the end of the century (2080–2099). This will lead to more heatwaves, higher energy demand and increased health risks.



Temperature projection (low emissions scenario) Photo: [AdaptNSW](#)

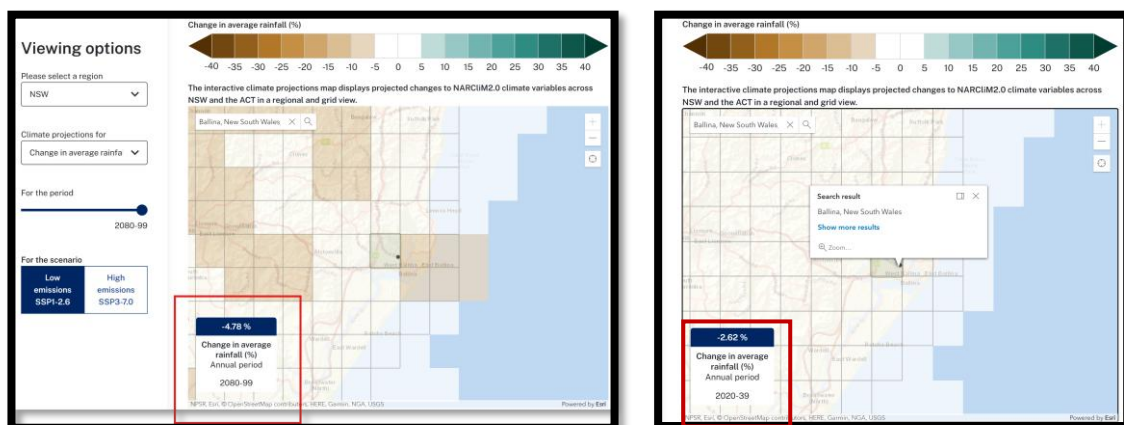
Rainfall trends and projections



Annual rainfall variability. Photo: Bureau of methodology

Ballina experiences considerable variability in rainfall, with both wet and dry years. Climate projections suggest average rainfall is expected to decrease slightly. Some years record extremely high rainfall (above 2500–2900 mm), while others drop to much lower levels (around 1000–1500mm). This significant variation highlights that Ballina's climate is naturally variable, affected by factors such as coastal weather systems and events like [El Niño](#) and [La Niña](#).

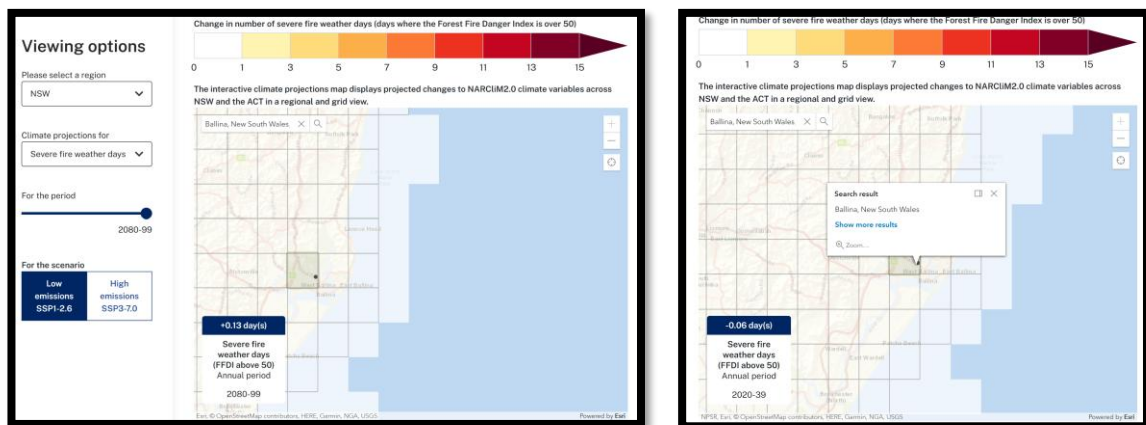
Projected rainfall change is -2.62% (2020–2039) and -4.78% (2080–2099). Even with this decrease, rainfall intensity is expected to increase, raising flood risk.



Rainfall projection (Low emission Scenario). Photo: AdaptNSW

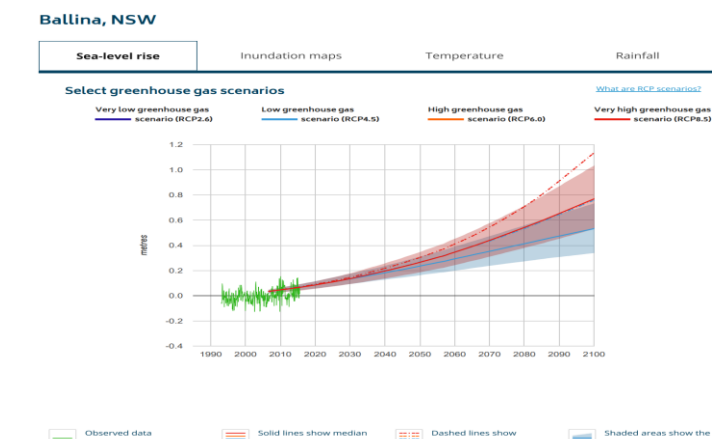
Fire weather risk

Severe fire weather days (FFDI > 50) exhibit minimal alterations under low emissions: -0.06 days (short-term) and $+0.13$ days (long-term). Although they are minor, risks could grow in severity with the rise of emissions scenarios. An increase in severe fire weather days (high temperature, low humidity, strong winds) can lead to greater chance of bushfires in nearby areas, fires spreading more quickly and becoming difficult to control. Even minor escalations in fire weather can elevate the overall risk, particularly during dry days.



Fire weather projection. Photo: AdaptNSW

Sea-level rise



Sea-level rise projections. Photo: CoastAdapt

Ballina faces a significant danger from rising sea levels. Projections show increases of $\sim 0.13\text{m}$ by 2030, $\sim 0.22\text{--}0.27\text{m}$ by 2050, and up to $\sim 0.66\text{m}$ by 2100. Due to elevated sea levels, waves can advance further inland, resulting in beach erosion, dune damage, and greater risk to coastal properties. Storms will push water further inland than before, resulting in deeper flooding, greater damage to infrastructure, and heightened risks to homes and roads.

How these projections are made

Data from the [Bureau of Meteorology \(BOM\)](#) were used in this analysis, along with future projections derived from NARCIIM2.0 and [IPCC](#) climate models. These models replicate the interactions of the atmosphere, oceans and land systems to forecast future climate conditions. Because of differences in future emissions, model variations and natural climate variability, there is always some uncertainty in climate projections. Nonetheless, every model reliably

demonstrates the same general tendency: temperatures are rising, sea levels are ascending, and climate risks are growing.

What this means for Ballina's future

Ballina is already experiencing the initial effects of climate change, including rising temperatures, fluctuating rainfall patterns, heightened environmental risks and an increase in sea levels. Although short-term changes may appear minor, their cumulative impact will grow over time. The town will likely face increased heat, flooding, erosion and strain on its infrastructure in the future. Nonetheless, with adequate planning and measures taken, the effects can still be mitigated. To create a resilient community, it is crucial to increase public awareness of and understanding of these risks.